INTERNNOVA PYTHON INTERNSHIP

WEEK 1 ASSIGNMENT REPORT

PYTHON FUNDAMENTALS

Submitted By: Ajay Garshakurthi

Student ID: **INST26PP080040**

E-mail: ajaygarshakurthi9@gmail.com

Mobile Number: 7013080514

College Name: Malla Reddy College of Engineering

Internship Organization: INTERNNOVA

Domain: Python Programming

Week: 01

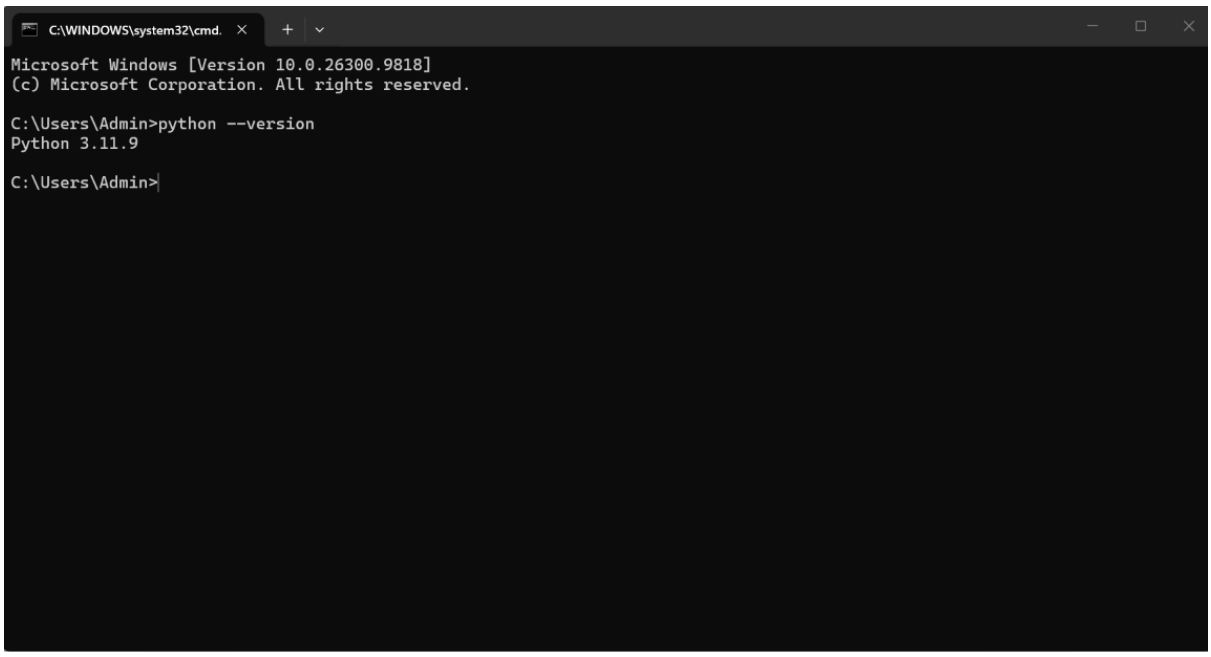
Introduction:

Python is a high-level, interpreted and general-purpose programming language. It is widely used in software development, data science, artificial intelligence, machine learning and automation.

Objectives:

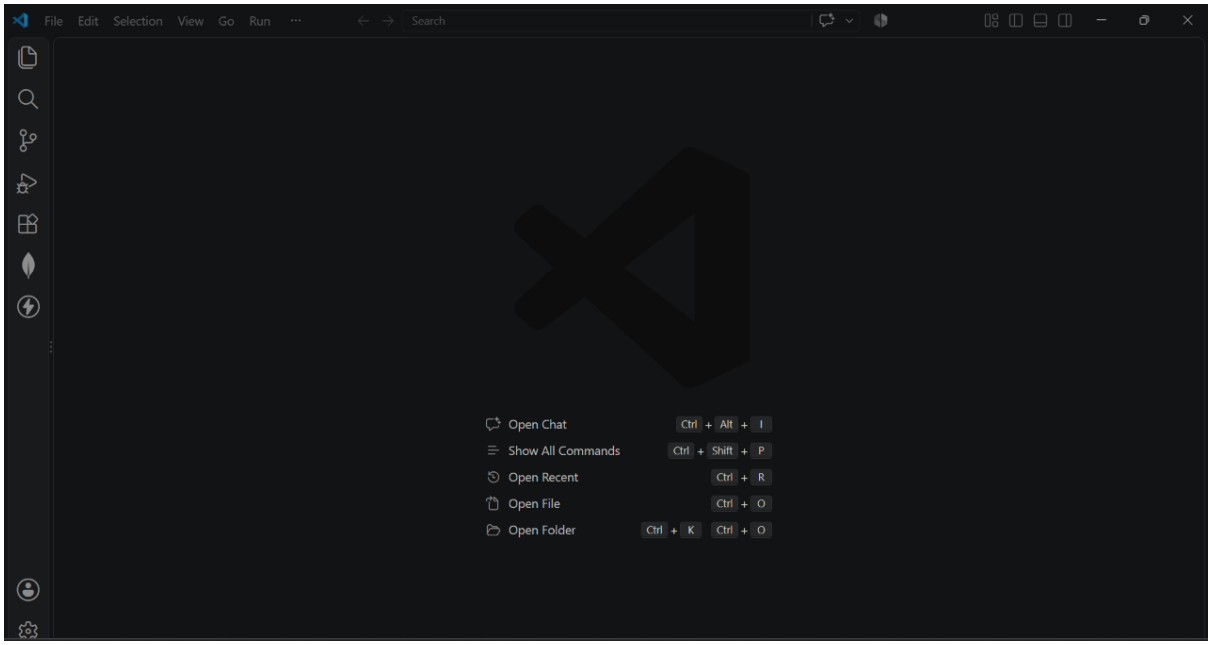
- To install Python and configure Visual Studio Code.
- To understand variables and different data types.
- To learn input and output functions.
- To understand arithmetic operators.
- To build a Mini Python program

TASK-1: Python Installation & Setup:



Python was successfully installed and configured, and the first Python program was executed.

Installation of VS Code:



VS Code installed Successfully, It's a lightweight source code editor that provides an environment for writing, editing and executing Python programs. It supports various programming languages and extensions, making it suitable for Python development.

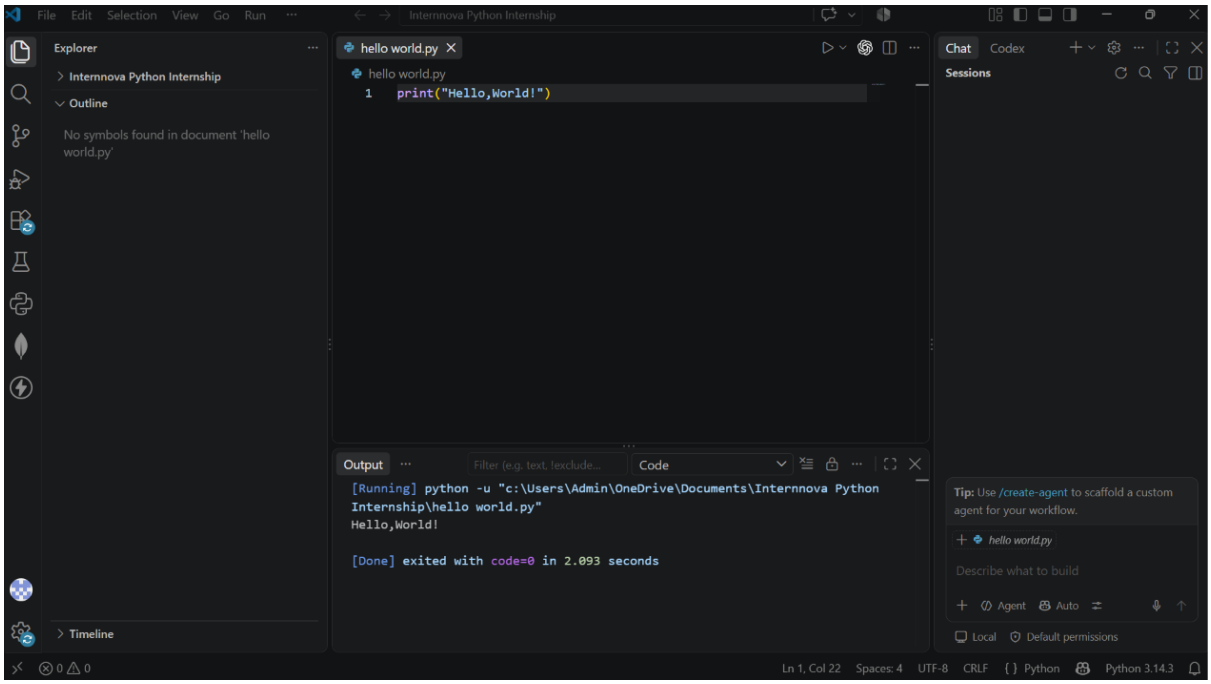
Python Extension Download in VS Code:



The Python extension developed by Microsoft was successfully installed in Visual Studio Code. This extension provides essential features such as Python code execution, syntax highlighting, debugging and IntelliSense. It helps configure the Python interpreter and makes Python programming easier

Hello World Program:

Aim: To write and execute the first Python program using VS Code.



Explanation:

The print() function is used to display messages on the screen. In this program, two messages are displayed to verify that Python is successfully configured and working in VS Code.

Result:

The Hello World program was successfully executed in VS Code, confirming that the Python installation and environment setup were successful.

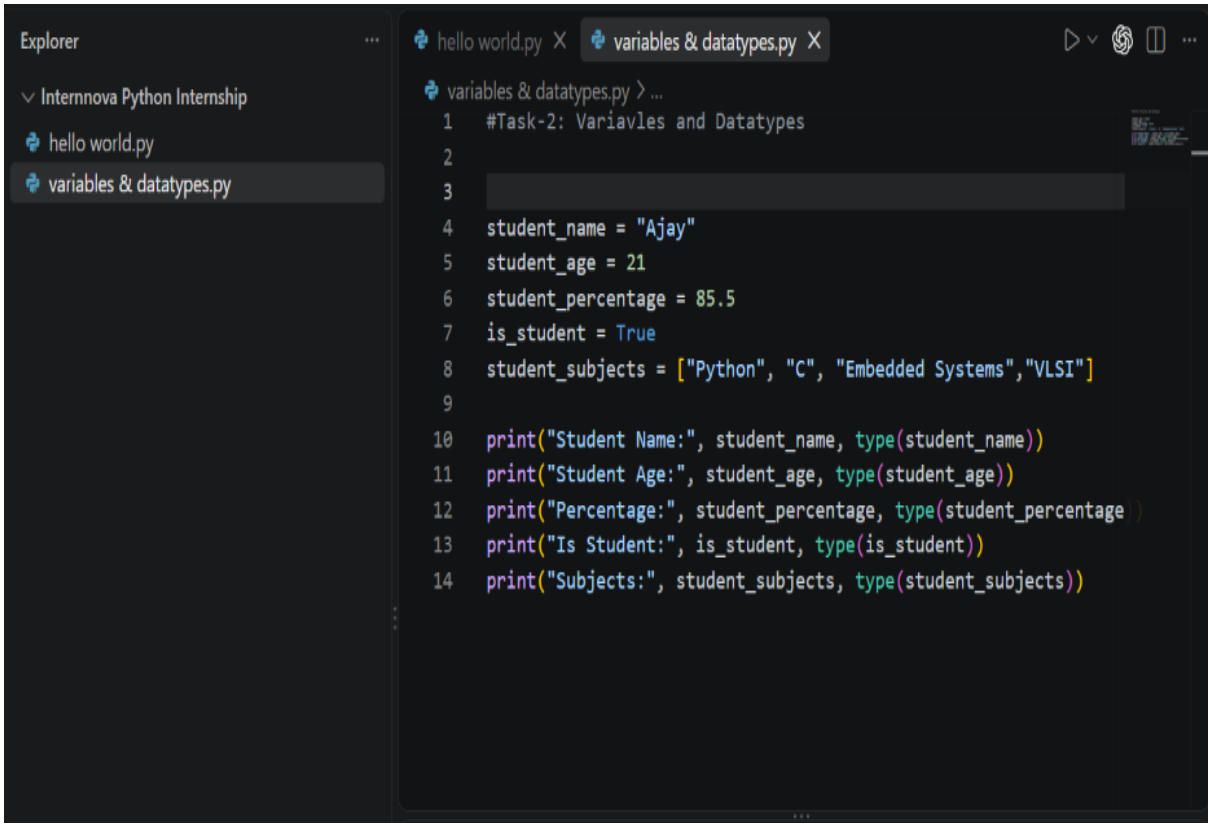
Task-2: Variables & Data Types:

Aim: To declare variables of different data types and display their values and types using the type() function.

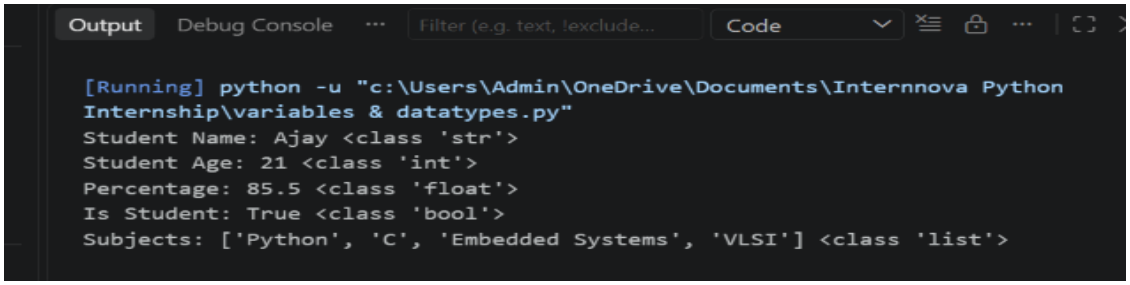
Concepts Used:

- Integer
- Float
- String
- Boolean
- List
- Tuple
- Dictionary

Code:



Output:



Explanation:

The program demonstrates different Python data types. The type() function identifies the data type of each variable, while the print() function displays the values.

Result:

Successfully created variables of different data types and displayed their values and types.

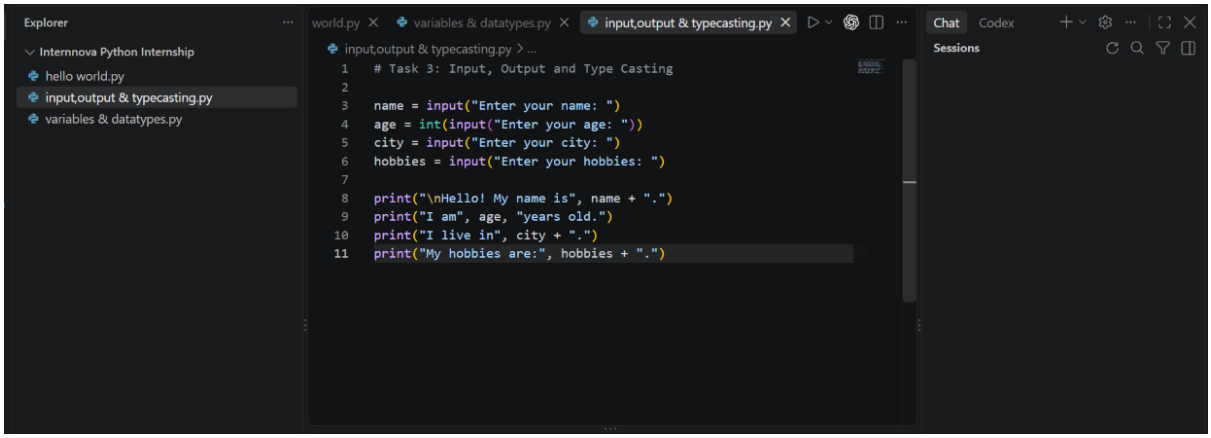
Task 3 – Input, Output & Type Casting:

Aim: To accept user information and convert the age into an integer using type casting.

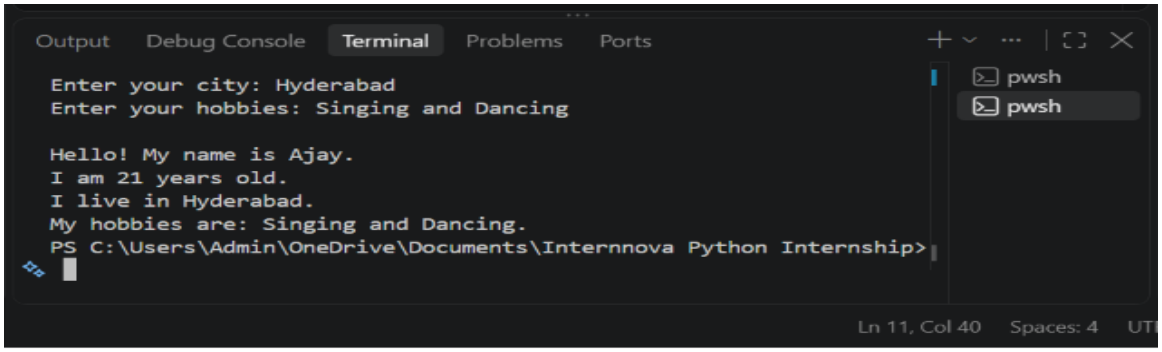
Concepts Used:

- input()
- print()
- int()

CODE:



Output:



Explanation:

The program accepts the user's name, age and city. The age is initially received as a string and converted into an integer using int(). Finally, the introduction is displayed using the print() function.

Result:

Successfully implemented user input, output and type casting.

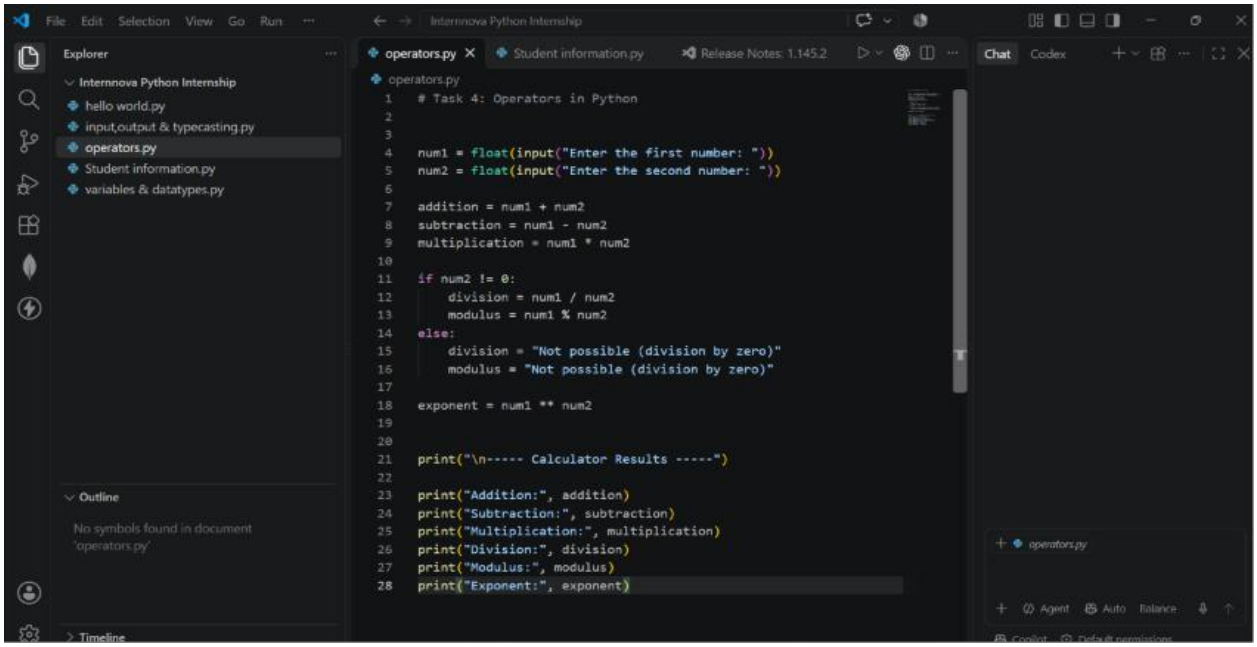
Task 4 – Operators in Python:

Aim: To perform arithmetic operations using Python operators**.**

Operators Used:

Operator	Symbol
Addition	+
Subtraction	-
Multiplication	*
Division	/
Modulus	%
Exponent	**

Code:



```python
# Task 4: Operators in Python


num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))

addition = num1 + num2
subtraction = num1 - num2
multiplication = num1 * num2

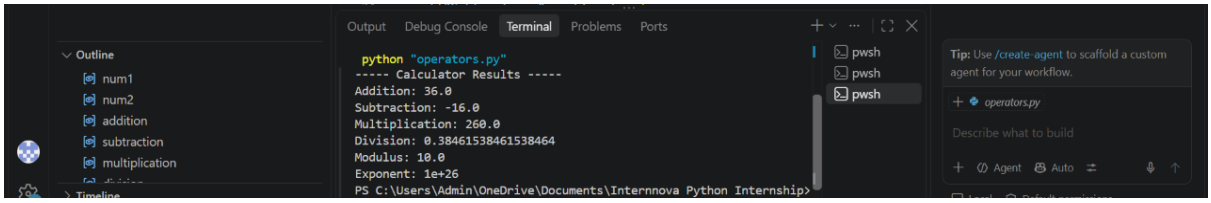
if num2 != 0:
    division = num1 / num2
    modulus = num1 % num2
else:
    division = "Not possible (division by zero)"
    modulus = "Not possible (division by zero)"

exponent = num1 ** num2


print("\n----- Calculator Results -----")

print("Addition:", addition)
print("Subtraction:", subtraction)
print("Multiplication:", multiplication)
print("Division:", division)
print("Modulus:", modulus)
print("Exponent:", exponent)
```

Output:



```
python "operators.py"
----- Calculator Results -----
Addition: 36.0
Subtraction: -16.0
Multiplication: 260.0
Division: 0.38461538461538464
Modulus: 10.0
Exponent: 1e+26
PS C:\Users\Admin\OneDrive\Documents\Internnova Python Internship>
```

Explanation:

The program accepts two numbers from the user and performs six arithmetic operations. The results are displayed using the print() function. A condition is also used to handle division by zero.

Result:

Successfully implemented all six arithmetic operators in Python.

Task 5 – Mini Python Program:

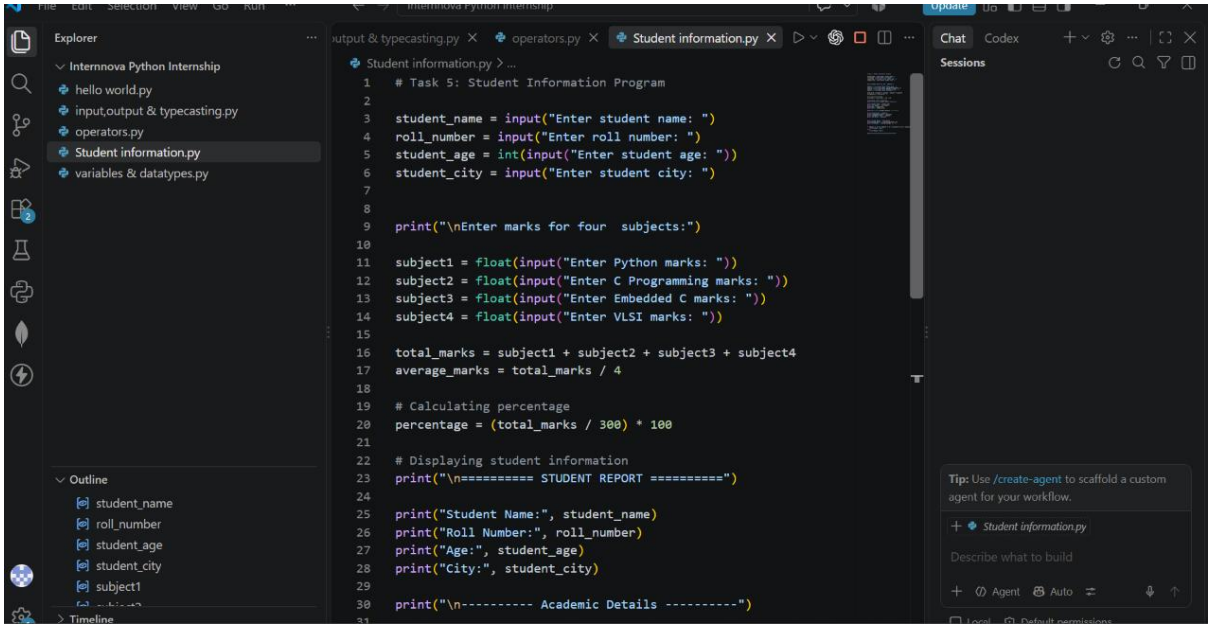
Topic: Student Information Program

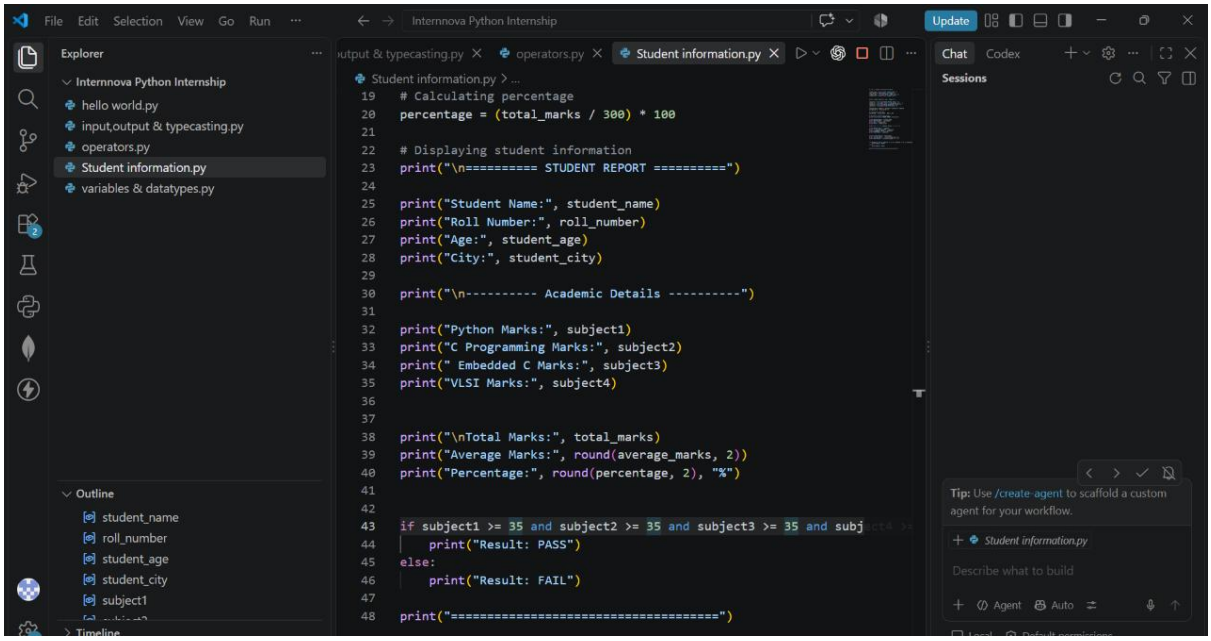
Aim: To develop a student information system using variables, input, operators and type casting.

Concepts Used:

- Variables
- User input
- Type casting
- Arithmetic operators
- Conditional statements
- Output formatting

Code:





```
# Calculating percentage
percentage = (total_marks / 300) * 100

# Displaying student information
print("\n========== STUDENT REPORT ==========")

print("Student Name:", student_name)
print("Roll Number:", roll_number)
print("Age:", student_age)
print("City:", student_city)

print("\n---------- Academic Details ----------")

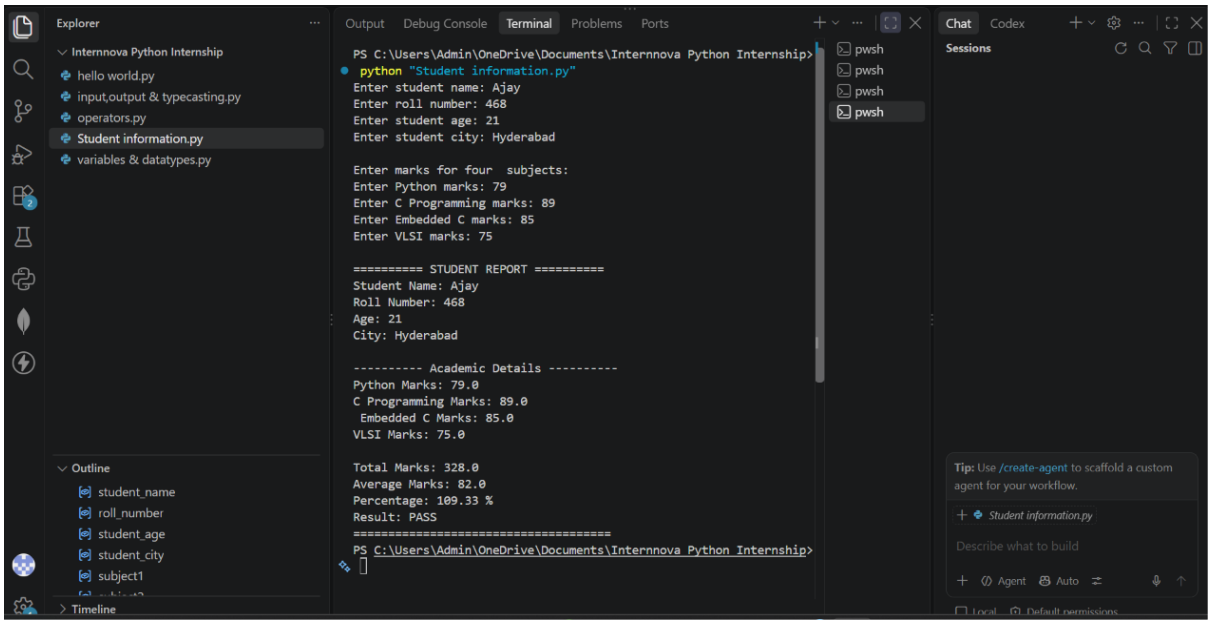
print("Python Marks:", subject1)
print("C Programming Marks:", subject2)
print(" Embedded C Marks:", subject3)
print("VLSI Marks:", subject4)


print("\nTotal Marks:", total_marks)
print("Average Marks:", round(average_marks, 2))
print("Percentage:", round(percentage, 2), "%")


if subject1 >= 35 and subject2 >= 35 and subject3 >= 35 and subj
    print("Result: PASS")
else:
    print("Result: FAIL")

print("====================================")
```

Output:



```
PS C:\Users\Admin\OneDrive\Documents\Internnova Python Internship>
 python "Student information.py"
Enter student name: Ajay
Enter roll number: 468
Enter student age: 21
Enter student city: Hyderabad

Enter marks for four  subjects:
Enter Python marks: 79
Enter C Programming marks: 89
Enter Embedded C marks: 85
Enter VLSI marks: 75

========== STUDENT REPORT ==========
Student Name: Ajay
Roll Number: 468
Age: 21
City: Hyderabad

---------- Academic Details ----------
Python Marks: 79.0
C Programming Marks: 89.0
 Embedded C Marks: 85.0
VLSI Marks: 75.0

Total Marks: 328.0
Average Marks: 82.0
Percentage: 109.33 %
Result: PASS
====================================
PS C:\Users\Admin\OneDrive\Documents\Internnova Python Internship>
```

Explanation:

The program collects student details, including name, roll number, age and city. It accepts marks in three subjects and calculates the total marks, average and percentage. It also checks whether the student has passed or failed.

Result:

Successfully developed a Student Information Program using Python fundamentals.

Conclusion:

Through this assignment, I gained practical knowledge of Python programming fundamentals. I learned how to install and configure Python, declare variables, work with different data types, accept user input, perform type casting and use arithmetic operators.